

Literature Status Note: Extreme Points of Lipschitz-Free Spaces

fa_banach_001, lane 19

2026-06-20

Status. `literature_already_answered`. This is a literature-status packet, not an original proof packet.

Source/trigger. Ch. Cobollo and A. Peris, *On disjoint dynamical properties and Lipschitz-free spaces*, arXiv:2405.12626v2. The extracted signal occurs in Section 4.1, “Lipschitz-free working preliminaries”, around source line 635: the authors recall the area-level conjecture that every extreme point in a Lipschitz-free space should be a molecule. This is background context rather than a numbered new problem of that paper.

Answer source. R. J. Aliaga, E. Pernecka, and R. J. Smith, *A solution to the extreme point problem and other applications of Choquet theory to Lipschitz-free spaces*, arXiv:2412.04312. The abstract states that all extreme points of the unit ball of Lipschitz-free spaces are elementary molecules. Section 4 is called “The extreme point problem”; it records the historical problem and then gives Theorem 4.1.

Answer theorem. Theorem 4.1 of arXiv:2412.04312 states that if M is a complete metric space, then

$$\text{ext } B_{\mathcal{F}(M)} \subset \text{Mol}(M).$$

More precisely, the extreme points of $B_{\mathcal{F}(M)}$ are exactly the molecules

$$m_{xy} = \frac{\delta_x - \delta_y}{d(x, y)}$$

with $x \neq y \in M$ such that

$$d(x, p) + d(p, y) > d(x, y) \quad \text{for every } p \in M \setminus \{x, y\}.$$

Since Lipschitz-free spaces do not change under completion, this supplies the usual formulation for arbitrary metric spaces after replacing M by its completion.

Why this answers the source signal. The source signal from arXiv:2405.12626 was exactly the molecule conjecture for extreme points of Lipschitz-free spaces. The supporting paper explicitly announces a solution to that extreme-point problem and provides the final classification. Thus this target should not be attacked as a fresh open problem in this run.

Scope limitation. Only the background extreme-point conjecture is cleared. This packet does not claim to answer any separate disjoint dynamical property question from arXiv:2405.12626.

References

- [1] Ch. Cobollo and A. Peris, *On disjoint dynamical properties and Lipschitz-free spaces*, arXiv:2405.12626v2, 2024.
- [2] R. J. Aliaga, E. Pernecka, and R. J. Smith, *A solution to the extreme point problem and other applications of Choquet theory to Lipschitz-free spaces*, arXiv:2412.04312, 2024.